

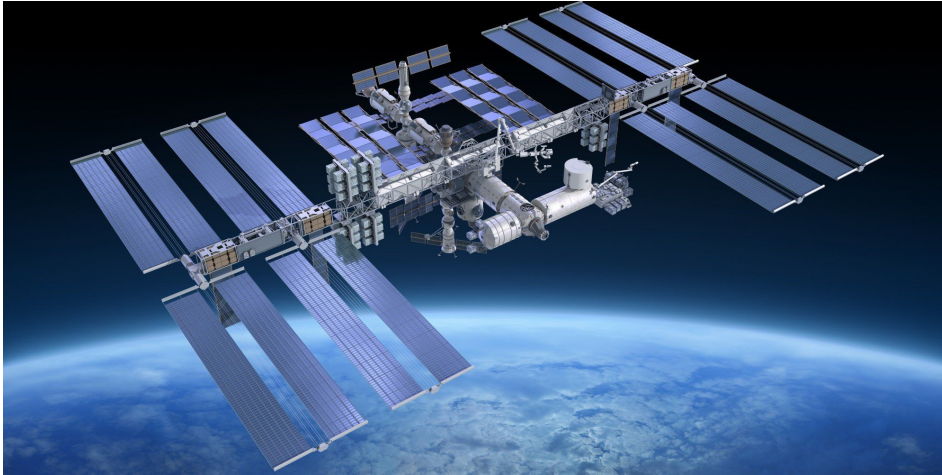
Let's learn about **satellites!**

Space mission and **satellite** design course

Spacecraft electrical power system

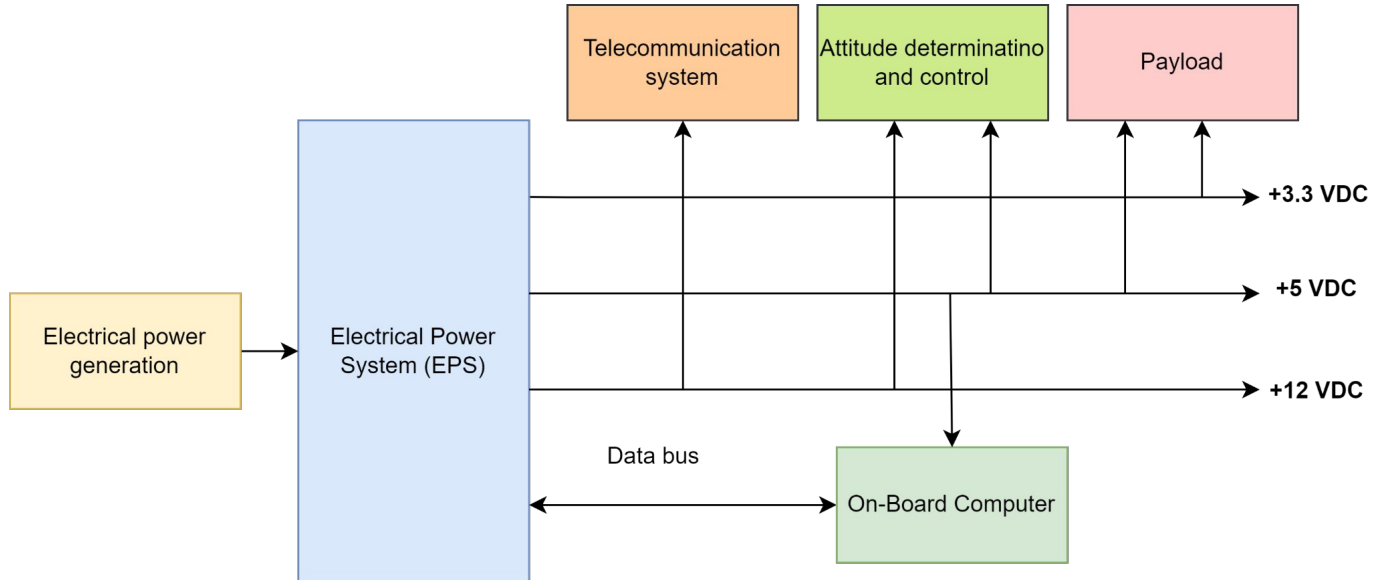
Electrical Power System

Satellites need a system to **generate, dispense** and **control** electrical power.



High level system view

Example of a simple electrical power system



High level system view

Satellite Electrical Power System (**EPS**) is composed mainly by:

- Power generation systems
- Energy storage systems
- Power management and distribution systems

Power generation

We have different **sources** of power in space:

- Solar panels
- Radioisotope Thermoelectric Generator (RTG)
- Batteries
- Fuel cells
- Nuclear reactor

Solar panels

Solar panels is the most common and simplest way to generate electrical power. It can be used up to **Jupiter**. After this planet the solar power is too low to be used.

Solar panels are composed by **solar cells** which generate electrical power through the **photovoltaic effect**.

$$P = E \varepsilon A \cos \theta$$

Solar panels

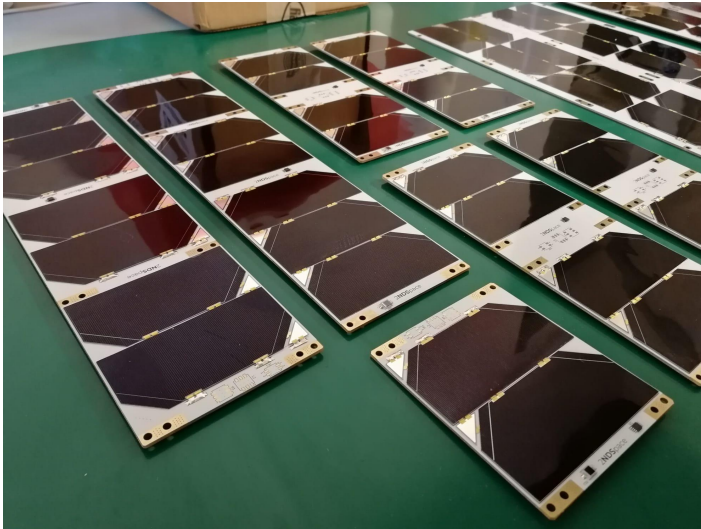
Solar-cell efficiency refers to the portion of energy in the form of sunlight that can be converted via photovoltaics into electricity by the solar cell.

We have mainly two types of solar cells:

1. **Single junction** with efficiency between **15%** and **19%**. They are the cheapest
2. **Triple junction** cells can achieve **29%** to **31%** of efficiency with a significant higher cost.

Solar panels

Triple junction solar cells are the most used since we want to maximize the power generation with respect to mass and volume used.



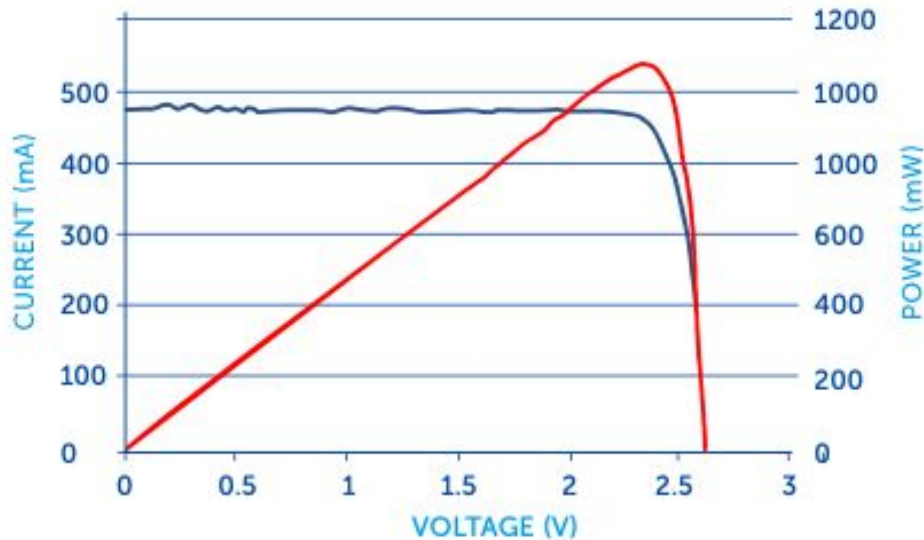
CESI CTJ30 solar cell

Solar panels

Example of a space grade solar cell data from italian provider CESI

Ref. **CTJ-30**

<https://www.cesi.it/app/uploads/2020/03/Datasheet-CTJ30-1.pdf>

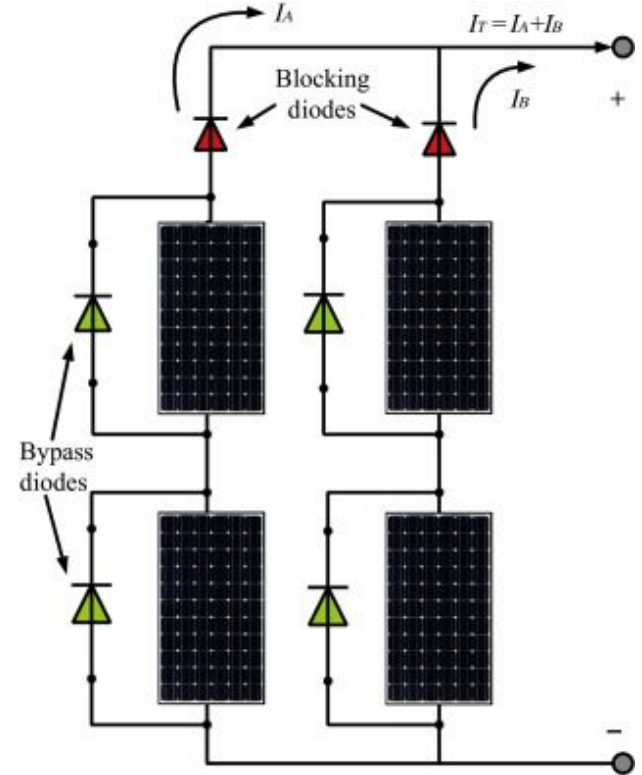


Area (cm ²)	I _{sc} (mA)	V _{oc} (V)	I _m (mA)	V _m (V)	P _{max} (W)	Eff (%)
26.5	473	2.6	455	2.32	1.05	29.0
30.15	538	2.61	517	2.33	1.20	29.0

Solar panels

Solar cells can be placed in series or in parallel. The series panel defines the maximum voltage while the parallel cells defines the maximum current that can be generated.

Blocking diodes and **bypass diodes** are critical elements in space solar panels



Solar panels

In order to maximize power generation it is necessary to:

- Pilot the panel voltage with proper **MPPT** control system
- Increase the solar panels **Area** as much as possible
- Maintain the panels **oriented** towards the sun
- Exploit cells with the highest **efficiency**

Radioisotope thermal generator

This method of electrical power generation takes the heat generated by decaying isotopes (Plutonium 238) and converts it to electrical power.

Extremely expensive system and extremely low efficiencies make this method limited to space agencies usage.

It is used when:

1. Mission is too **far** from the Sun
2. Mission requires **stable** electrical power output regardless of the spacecraft attitude and ambient conditions

Radioisotope thermal generator

Exploiting the **Seebeck** effect we are able to generate electrical power from two semiconductors with different temperatures. This method has a typical **efficiency** of around **2%** which is extremely low, making the **RTG** a very expensive choice.

For instance, if we need constant **300W** electrical power for our mission we will have to manage **15,000W** of heat. Hence we will have to create a very complex cooling system which will depend on the type of mission.

Batteries

Batteries can be used as **primary** source of energy or **secondary** source of energy.

If the mission length is in days batteries can represent a solid choice. For example launchers have batteries as their primary source of energy.

Batteries now are either Lithium-Ion or Lithium polymer.

The energy density of the batteries are around **250** Wh/kg.

Fuel cells

Fuel cells has a higher energy density than batteries but they are more complex, hence more expensive and massive.

They use hydrogen and oxygen to generate electricity and water as final waste.

It is not a common choice but the most famous example is the **Space Shuttle** which relied in this type of energy generation to complete its missions.

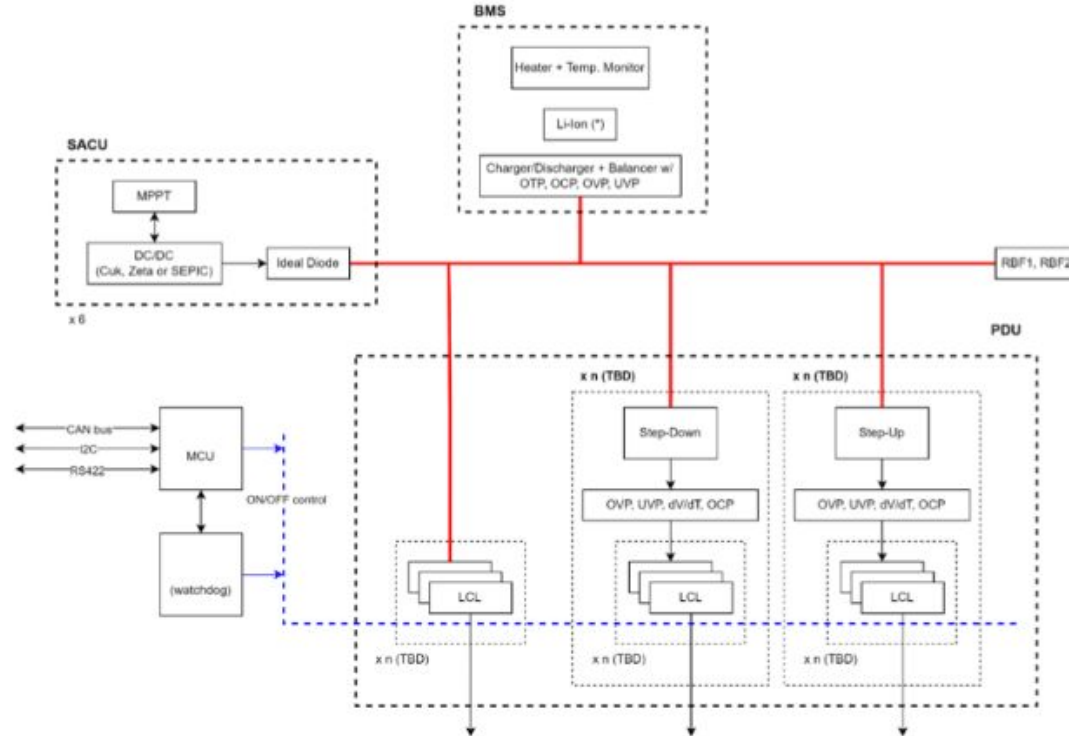
Nuclear reactor

Nuclear reactors for satellite are a smaller redesign of a standard nuclear power plant. (useful if we want more than 10kW of power)

There is not much history with this field due to the extreme **complexity** and **danger** during construction and launch.

It will be more probably used for a **human base** on the Moon or Mars due to the extremely high density power and reliability. The wasted heat can also be used for heating up the base itself.

Power management system



Solar panel design

99% of the time you will use solar panels as primary source of energy so let's discuss how you will do the design of the system.

The Sun energy can be measured in W/m^2 and it depends on the distance between the satellite and the Sun, here are some examples:

- **Venus** distance: 2615 W/m^2
- **Earth** distance: 1367 W/m^2
- **Mars** distance: 592 W/m^2
- **Jupiter** distance: 50 W/m^2

Solar panel design

We would like to design a 2U mission in low Earth orbit with 500 km altitude, which will consume an average power of **6W**.

Let's start by using triple junction solar cells that have **30%** efficiency. The Energy that can be gathered is:

$$P = 1367 \eta = 1367 * 0.3 = 410 \text{ W/m}^2$$

Solar panel design

Due to some inefficiencies the previous number will be reduced typically by around 20%.

$$P = 410 * 0.8 = 328 \text{ W/m}^2$$

The solar cells are directly proportional the cosine angle between the normal of the cell and the Sun direction. Imagining to have a medium angle between solar panels and Sun of 20 degree we arrive at the final number:

$$P = 328 * \cos(20\text{deg}) = 308 \text{ W/m}^2$$

Solar panel design

We know we need 6W hence we are able to know how much area of solar cells we will need:

$$A = \frac{P_{\text{sat}}}{P} = \frac{6}{308} = 0.0195 \text{ m}^2 = 195 \text{ cm}^2$$

A standard **CESI** solar cell is **30cm²** hence we will need at least **7 cell** to fulfill the mission.

Solar panel design

We will spend some time in eclipse during the orbit which means we will need a **secondary** source of energy: **batteries**.

The amount of time we spend in eclipse, for a circular orbit, is defined by the following formula:

$$T_e = T * \frac{\arcsin\left(\frac{6378}{6378 + h}\right)}{180}$$

Where **T** is the orbital period and h is the orbital **altitude** in km.

Solar panel design

In our case the orbit altitude is 500km and the orbit period is 95 minutes:

$$T_e = 95 \text{ min} * \frac{\text{asin}\left(\frac{6378}{6378 + 500}\right) [\text{deg}]}{180} = 36 \text{ min}$$

One important parameter for the battery design is the **Depth of Discharge**.

We cannot use 100% of battery capacity otherwise the life length will decrease significantly and increase the failure risk. A typical DoD is 20%, which means that the battery will always work with >80% capacity

Solar panel design

The formula for the battery sizing is the following:

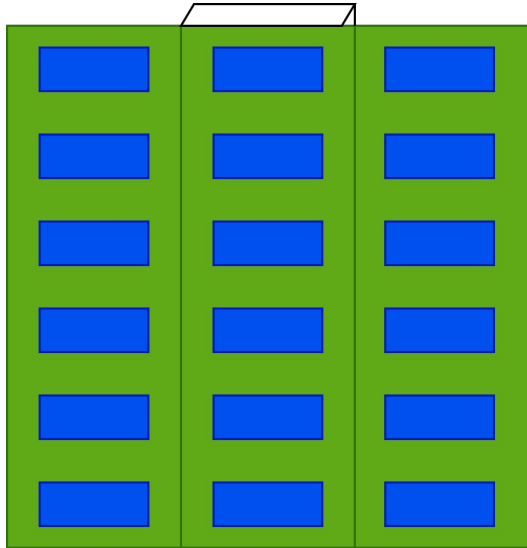
$$C = \frac{T_e * P_e}{DoD}$$

In our case the time spent in eclipse is 35 min and the power consumed in eclipse is always 6W:

$$C = \frac{6W * 0.58 h}{0.2} = 17.4 Wh$$

Solar panel design

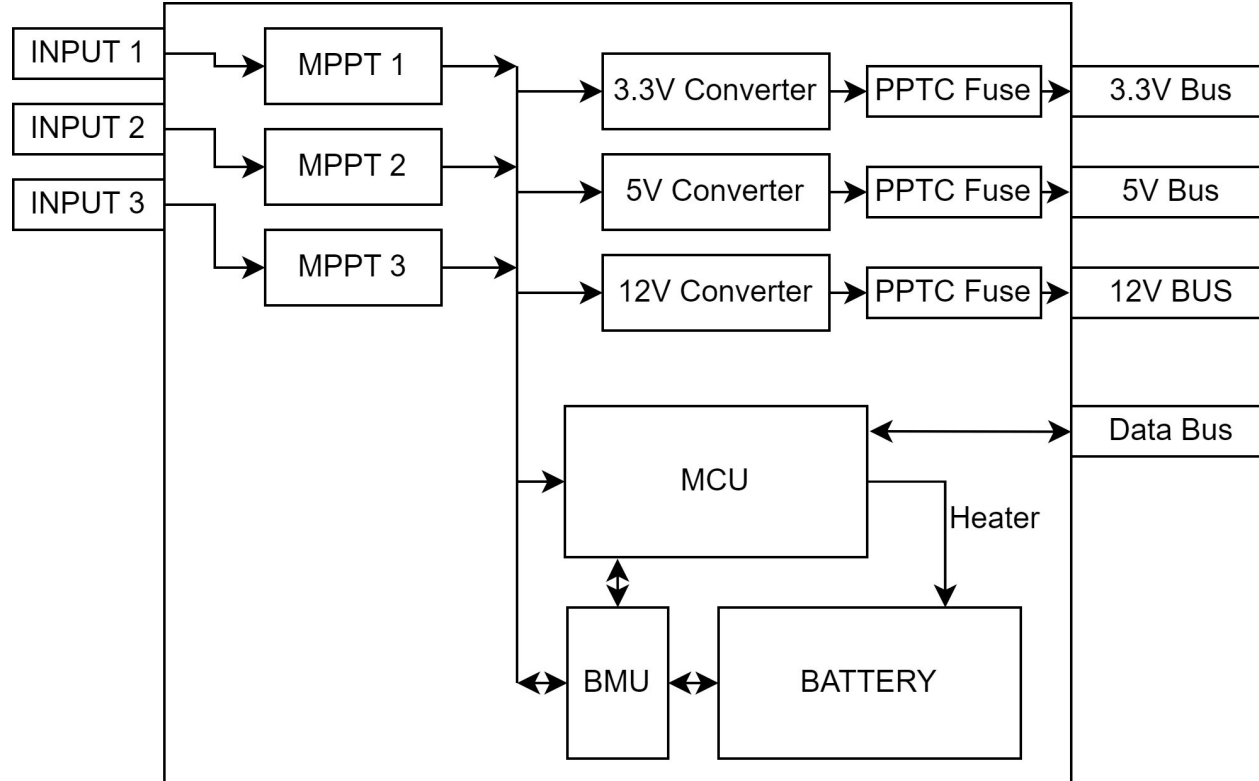
Our satellite could look like this



The battery pack will be composed by 2 18650A Lithium-ion battery



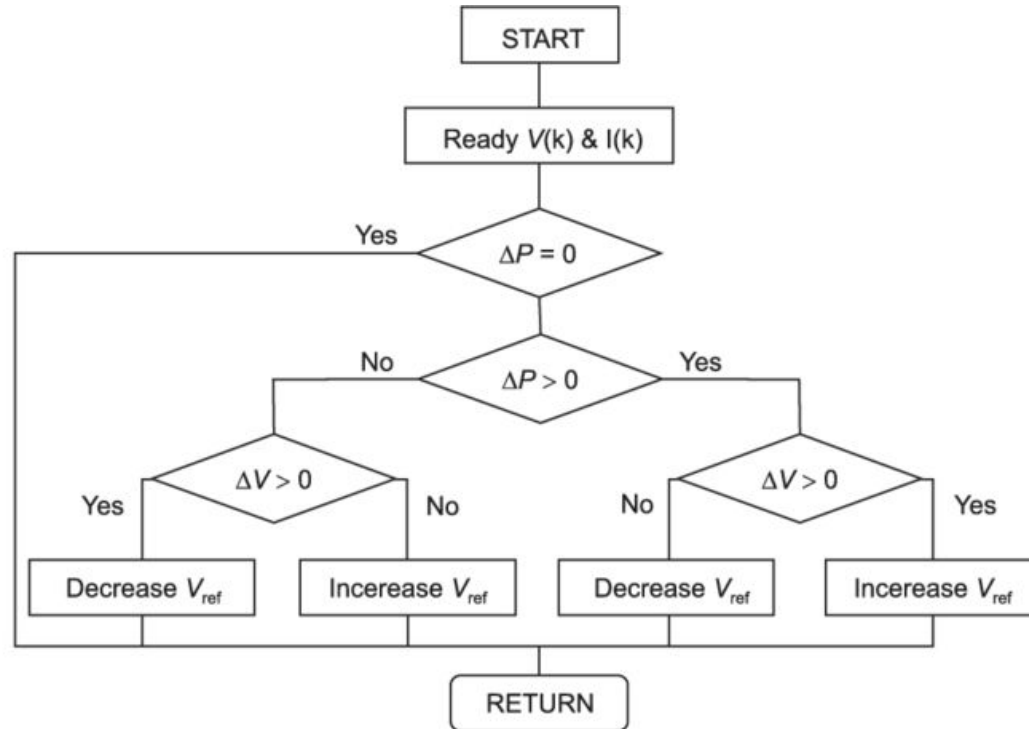
Electrical Power System design



MPPT

The Maximum Power Point Tracking must always and continuously find the maximum power point in order to extract the **maximum** energy possible.

MPPT - Algorithm



BMU

The battery management unit is in charge of keeping the batteries in **healthy** state during charging or discharging. It will protect the batteries from **overcharge** and **over discharge** and will **balance** the charge of batteries if they are more than one.

MCU

The micro-controller unit **supervises** that everything is running smoothly and is in charge of communicating to the on-board computer all the **telemetry**.

It can take **commands** to shut off or open certain power buses. If the batteries are too low in temperature **battery heaters** are turn on in order to ensure to maintain a good temperature range for the batteries (which should be around room temperature)

DC-DC Converters

Battery voltages can not be used for all the applications. Different systems will need different voltages, the most common ones are: **3.3V**, **5V**, **12V**, **24V**. We have two type of converters:

- **Linear regulators:** cheap, simple but low efficiency
- **Switching regulators:** expensive, difficult but with high efficiency (in space, of course, we will use switching regulators)

DC-DC Converters

Two types of switching regulators:

- **Step up regulator:** takes lower voltage input and give an higher output voltage (e.g. from 7.4V to 5V)
- **Step down regulator:** takes higher voltage input and give a lower output voltage (e.g. from 7.4V to 12V)

PPTC Fuses

Fuses are used to protect the system from **overcurrent**.

Typical terrestrial fuses are made of a small wire that will burn and open the circuit when a certain amount of current goes through it (due to the Joule effect). After the fuse is triggered it will have to be **replaced**.

Polymeric Positive Temperature Coefficient fuses are made of a special polymer that after a certain temperature it will stop conducting and **open** the circuit. After some time (reset time) the polymer will **cool off** and start **conducting** again.

Questions?

Any questions?

Quiz - 1

Which is the scope of a batteries in a Power management unit

- A. Power generation
- B. Energy storage
- C. Power distribution
- D. Power storage

Quiz - 2

Select the most energy dense primary energy source:

- A. Fuel cells
- B. Nuclear reactor
- C. Radioisotope thermal generator
- D. Solar panels